JSP

1) JSP page to display “Hello World” using Expression Tag

```
<html>
<body>

<h2><%= "Hello World" %></h2>

</body>
</html>
```

---

2) JSP page to display current date and time using Scriptlet

```
<%@ page import="java.util.Date" %>

<html>
<body>

<h3>
<%
Date d = new Date();
out.println("Current Date and Time: " + d);
%>
</h3>

</body>
</html>
```

---

3) JSP page using Declarative Tag to declare a method

```
<html>
<body>

<%!
String greet(){
   return "Welcome to JSP Programming";
}
%>

<h2><%= greet() %></h2>

</body>
</html>
```

---

4) JSP page using request and response implicit objects

```
<html>
<body>

<%
String name = request.getParameter("name");

response.setContentType("text/html");

out.println("Hello " + name);
%>

</body>
</html>
```

Example URL:

http://localhost:8080/example.jsp?name=Pranjali

---

5) JSP page using session and application objects

```
<html>
<body>

<%
session.setAttribute("user","Pranjali");

application.setAttribute("college","Sangola College");

out.println("Session User: " + session.getAttribute("user"));
out.println("<br>Application: " + application.getAttribute("college"));
%>

</body>
</html>
```

---

6) JSP page demonstrating exception implicit object

error.jsp

```
<%@ page isErrorPage="true" %>

<html>
<body>

<h2>Error occurred</h2>

<%= exception.getMessage() %>
```

```
</body>
</html>
```

main.jsp

```
<%@ page errorPage="error.jsp" %>

<%
int a = 10;
int b = 0;

int c = a / b;
%>
```

---

7) JavaBean with name and age properties

Bean Class

```
public class Student {

   private String name;
   private int age;

   public void setName(String name){
      this.name = name;
   }

   public void setAge(int age){
      this.age = age;
   }

   public String getName(){
      return name;
   }

   public int getAge(){
      return age;
   }
}
```

JSP Page

```
<jsp:useBean id="s" class="Student" scope="page"/>

<jsp:setProperty name="s" property="name" value="Pranjali"/>
<jsp:setProperty name="s" property="age" value="21"/>

Name: <jsp:getProperty name="s" property="name"/><br>
Age: <jsp:getProperty name="s" property="age"/>
```

---

8) JSP page using <jsp:setProperty>

```
<jsp:useBean id="s" class="Student"/>

<jsp:setProperty name="s" property="name" value="Amit"/>
<jsp:setProperty name="s" property="age" value="22"/>

Name: <jsp:getProperty name="s" property="name"/><br>
Age: <jsp:getProperty name="s" property="age"/>
```

---

9) JSP page using JavaBean with form

HTML Form

```
<form action="bean.jsp">

Name: <input type="text" name="name"><br>
Age: <input type="text" name="age"><br>

<input type="submit" value="Submit">

</form>
```

bean.jsp

```
<jsp:useBean id="s" class="Student"/>

<jsp:setProperty name="s" property="*"/>

Name: <jsp:getProperty name="s" property="name"/><br>
Age: <jsp:getProperty name="s" property="age"/>
```

---

10) JSTL <c:if> example

```
<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c" %>

<c:set var="num" value="10"/>

<c:if test="${num > 0}">
Number is Positive
</c:if>
```

---

11) JSTL <c:choose> example

```
<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c" %>
```

```
<c:set var="marks" value="60"/>

<c:choose>

<c:when test="${marks >= 40}">
Pass
</c:when>

<c:otherwise>
Fail
</c:otherwise>

</c:choose>
```

---

12) JSTL <c:forEach> example

```
<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c" %>

<c:set var="arr" value="${['Java','Python','PHP','C++']}"/>

<c:forEach var="i" items="${arr}">

${i} <br>

</c:forEach>
```

---

13) JSTL <c:set> and <c:out>

```
<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c" %>

<c:set var="name" value="Pranjali"/>

Name: <c:out value="${name}"/>
```

---

14) JSTL <fmt:formatNumber> for currency

```
<%@ taglib uri="http://java.sun.com/jsp/jstl/fmt" prefix="fmt" %>

<fmt:formatNumber value="2500" type="currency"/>
```

---

15) JSTL <fmt:formatDate>

```
<%@ page import="java.util.Date" %>
<%@ taglib uri="http://java.sun.com/jsp/jstl/fmt" prefix="fmt" %>
```

```
<%
Date d = new Date();
request.setAttribute("date", d);
%>

<fmt:formatDate value="${date}" pattern="dd/MM/yyyy"/>
```

---

16) JSTL XML parsing

```
<%@ taglib uri="http://java.sun.com/jsp/jstl/xml" prefix="x" %>

<x:parse var="doc">

<students>
<student>
<name>Pranjali</name>
</student>
</students>

</x:parse>

<x:out select="$doc/students/student/name"/>
```

---

17) JSTL SQL <sql:query>

```
<%@ taglib uri="http://java.sun.com/jsp/jstl/sql" prefix="sql" %>

<sql:setDataSource var="db"
driver="com.mysql.jdbc.Driver"
url="jdbc:mysql://localhost:3306/testdb"
user="root"
password=""/>

<sql:query dataSource="${db}" var="rs">
SELECT * FROM student
</sql:query>

<table border="1">

<c:forEach var="row" items="${rs.rows}">

<tr>
<td>${row.id}</td>
<td>${row.name}</td>
</tr>
```

```
</c:forEach>

</table>
```

---

18) JSTL SQL <sql:update> insert record

```
<%@ taglib uri="http://java.sun.com/jsp/jstl/sql" prefix="sql" %>

<sql:setDataSource var="db"
driver="com.mysql.jdbc.Driver"
url="jdbc:mysql://localhost:3306/testdb"
user="root"
password=""/>

<sql:update dataSource="${db}">
INSERT INTO student VALUES (5,'Amit')
</sql:update>
```

Record Inserted Successfully

---

19) JSTL SQL <sql:param>

```
<%@ taglib uri="http://java.sun.com/jsp/jstl/sql" prefix="sql" %>

<sql:setDataSource var="db"
driver="com.mysql.jdbc.Driver"
url="jdbc:mysql://localhost:3306/testdb"
user="root"
password=""/>

<sql:update dataSource="${db}">
INSERT INTO student VALUES (?,?)
<sql:param value="6"/>
<sql:param value="Neha"/>
</sql:update>
```